

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2016-17

Paper No. GLB-101: Elementary Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem I/166

*Instructions: Attempt any **five** questions, selecting at least **two** questions each from Section A and Section B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Discuss in detail with diagrams the internal constitution of the Earth. [14]

Question 2. What do you understand by earthquakes? Discuss their causes, classification, and geographical distribution. [14]

Question 3. Define fold. What are the different parts of a fold? With the help of diagrams, give the classification of folds based on: [14]

- (a) arching direction
- (b) relative ages of beds
- (c) plunge of the hinge lines

Question 4. Write explanatory notes on the following: [2 × 7 = 14]

- (a) Unconformity
- (b) Forms of igneous rocks

SECTION — B

Question 5. Discuss the following: [2 × 7 = 14]

- (a) Convection currents in the Earth's core and production of the geomagnetic field
- (b) Geological time scale

Question 6. Write the salient differences between the following: [4 × 3.5 = 14]

- (a) Normal fault and reverse fault
- (b) Nappe and window
- (c) Outlier and inlier
- (d) Overlap and offlap

Question 7. Write short notes on **any four** of the following:

[4 × 3.5 = 14]

- (a) Dip and strike
- (b) Clinometer compass
- (c) Relative and absolute methods of age determination
- (d) Age of the Earth
- (e) Briefly discuss the origin of the atmosphere
- (f) Concordant and discordant bodies

SECTION — C (Compulsory)

Question 8. Write the answers to the following in two or three sentences, or fill in the blanks as the case may be:

[14]

- (a) What is anticlinorium? [1]
- (b) What is sill? [1]
- (c) What is master joint? [1]
- (d) What is thrust fault? [1]
- (e) What is fan fold? [1]
- (f) What type of dip does a dome structure have? [1]
- (g) What is outcrop? [1]
- (h) Earthquake magnitude is measured on the _____ scale and intensity is measured on the _____ scale. [2]
- (i) The most explosive volcanic eruption is of the _____ type and the least explosive is of the _____ type. [2]
- (j) The Big Bang happened _____ years ago and the Milky Way galaxy was formed _____ years ago. [2]
- (k) The volcanic belt around the Pacific Ocean is known as the _____. [1]

[Click here to visit our website](#)
[Click here to visit our contributor](#)
[Click here to visit our contributor](#)
